Memory Management, Virtual Memory and Deadlocks

Operating Systems - Units 3-4 - Memory and resources

Paging

Paging splits logical memory into fixed-size pages and physical memory into frames of the same size. A page table maps page numbers to frame numbers, which removes external fragmentation (internal fragmentation can remain in the last page).

```
logical address = (page number p, offset d)
physical address = frame(p) * page_size + d
page size 4 KB -> offset uses 12 bits
```

A Translation Lookaside Buffer (TLB) caches recent translations. Effective access time = hit ratio * (TLB + memory) + miss ratio * (TLB + 2 * memory).

Virtual memory and page replacement

Demand paging loads a page only when it is referenced. A page fault occurs when the page is not in memory and must be fetched from disk.

- FIFO: evict the oldest page; may show Belady's anomaly.
- Optimal: evict the page not needed for the longest time (theoretical benchmark).
- LRU: evict the least recently used page; a good practical approximation of optimal.
- Thrashing: the system spends more time paging than executing; reduce multiprogramming or use the working-set model.

Deadlocks

A deadlock is a set of processes each waiting for a resource held by another. It needs all four Coffman conditions simultaneously:

- Mutual exclusion - a resource is held by one process at a time.
- Hold and wait - a process holds resources while waiting for more.
- No preemption - resources cannot be forcibly taken.
- Circular wait - a cycle of processes each waiting for the next.

Handling deadlocks

- Prevention: break at least one Coffman condition, e.g. order resources to avoid circular wait.
- Avoidance: Banker's algorithm grants a request only if the system stays in a safe state.
- Detection and recovery: build a wait-for graph, then terminate or roll back processes.

NoteNest demo material - self-created for illustration, not an official publication.

> *Safe state means there is some order in which every process can finish using the currently available resources plus those released by earlier processes.*